

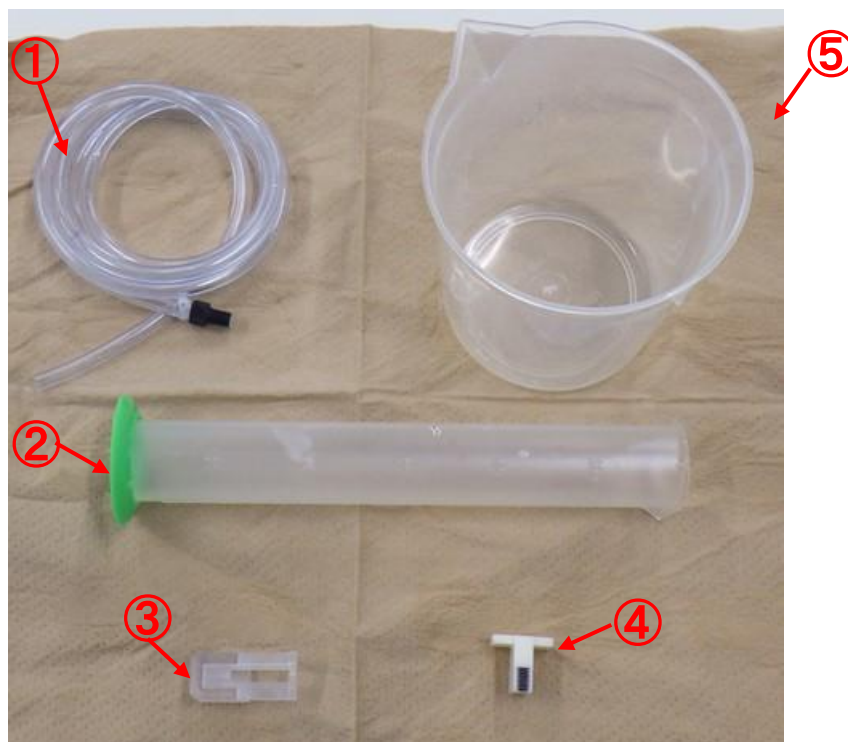
Adjustment procedure to CLEANING BATH S1 A (P/N 792-1324)

Applicable Serial Number : up to 2541-10 (Global ver.)

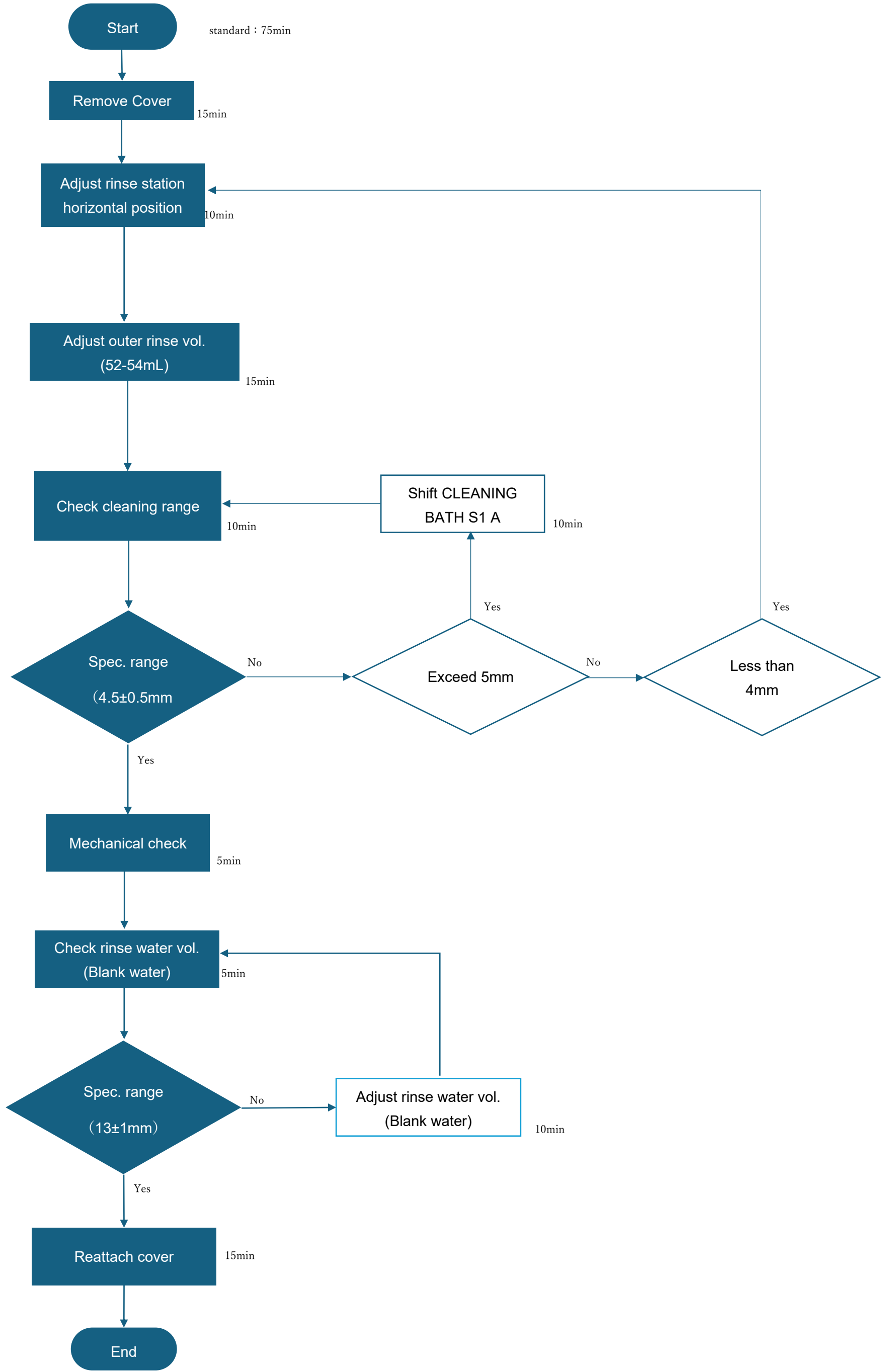
1.

1.1 Preparation

	P/N	Name	Pcs	Note
1	792-7752	ADJUTMENT TUBE SA	1	-
2	792-9022	GRADUATED CYLINDER 100	1	-
3	792-7687	ADJUSTMENT TOOL SA 1	1	-
4	792-6556	ADJUSTMENT TOOL SA 2	1	-
5	-	Container	1	Capable of receiving outside washing water.



1.2 Adjustment Workflow Overview

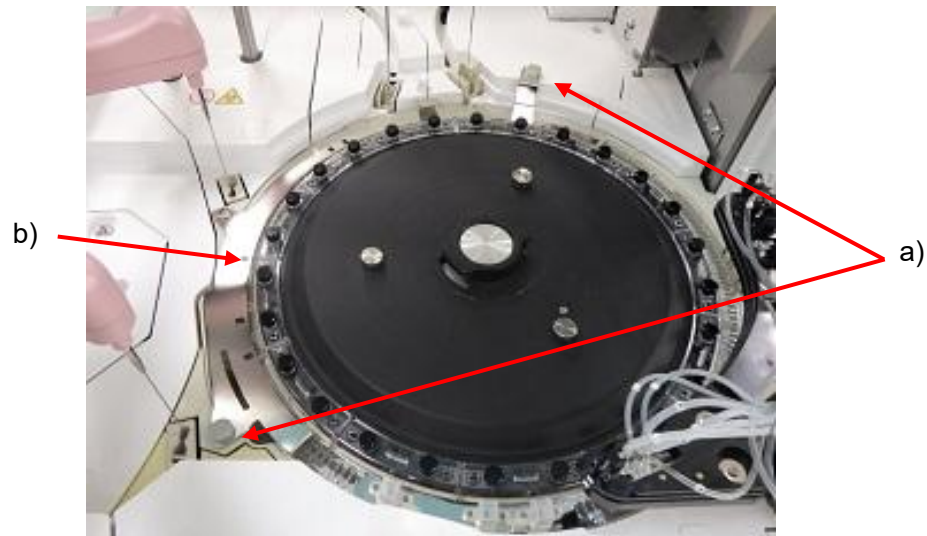


2. Remove Cover

2.1 Power off the instruments.

Be sure to wear gloves and eye protection with side shields during work.

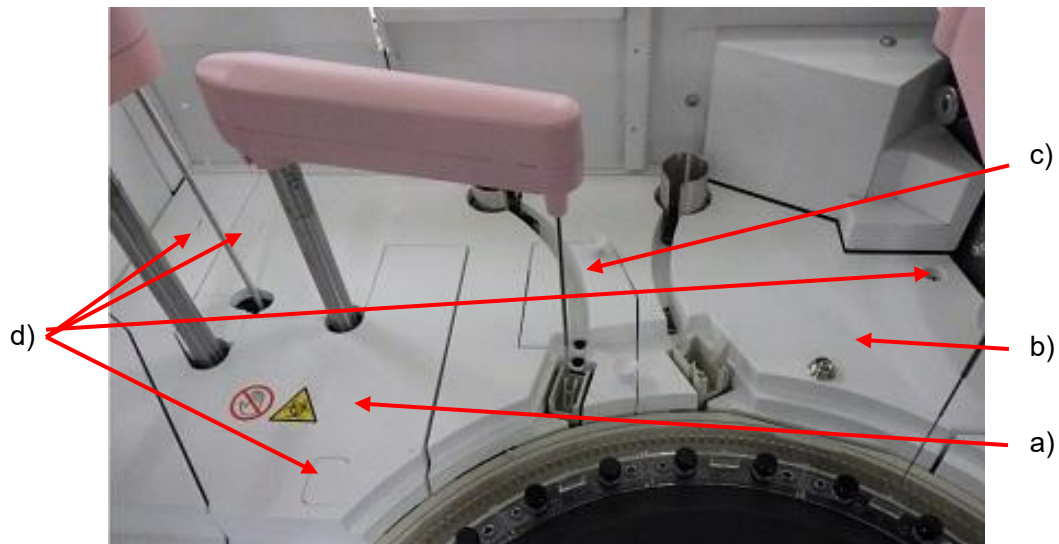
2.2 Remove the two nuts to remove the COVER RS CELL.



a) NUT

b) COVER RS CELL

2.3 Remove the COVER TB 4 ASSY, COVER TB 5 ASSY and COVER TB 7 (4 screws).



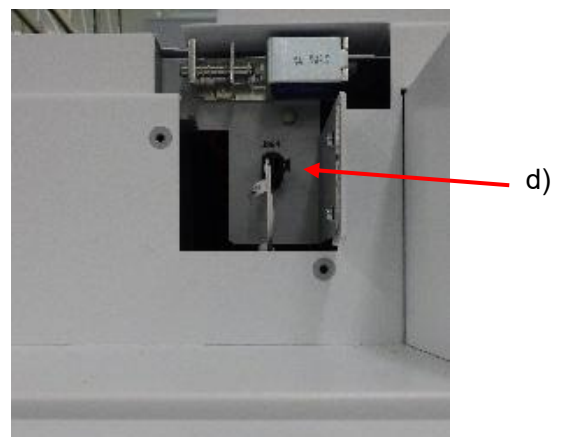
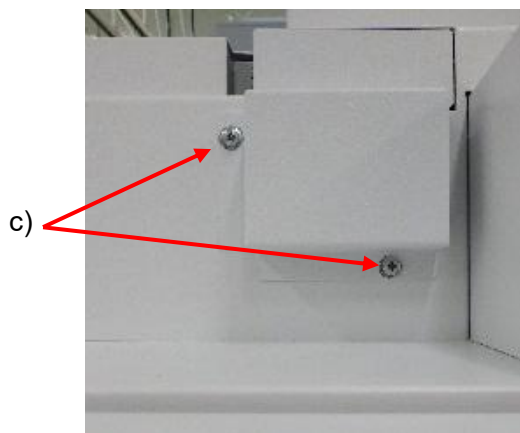
a) COVER TB 4 ASSY

b) COVER TB 5 ASSY

c) COVER TB 7

d) Screw

2.4 Remove the COVER HAND RA ASSY and COVER REAR ASSY (8 screws, 1 connector).

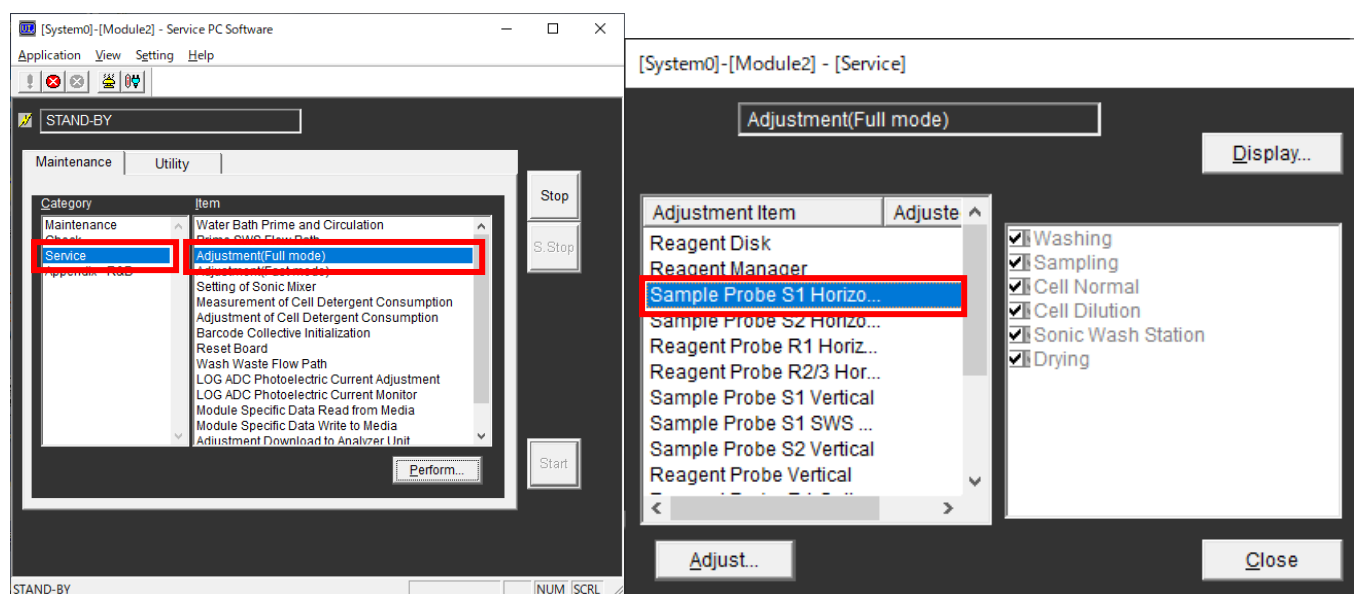


- a) COVER HAND RA ASSY
- b) COVER REAR ASSY
- c) Screw
- d) Connector

3. Turn on the instrument
 - 3.1. Turn on the instrument
 - 3.2. Check and adjustment the water supply pressure.
 V2S1: 50~60kPa (adjustment by visual and manually)
 V2S2: 60kPa (adjustment by SSW)
 - 3.3. Turn on maintenance mode by maintenance key.



4. Adjustment of location for S1 Rinse station.
 - 4.1. Perform adjustment item of Sample Probe S1 Horizontal by SSW.



- 4.2. Press the Next button to move the S1 probe out of Rinse Station.
- 4.3. Put ADJUSTMENT TOOL SA 1(P/N 792-7687) on S1 Rinse station.

ADJUSTMENT TOOL SA 1



4.4. Press the NEXT button until the S1 probe reaches Rinse Station.

[System0] - [Module2] - [Service]

Washing

Adjustment

X-Y

Z

Position

Y

Adjusted -7

Current -7

Action

Accept

Accept & Next

Set Up

Back

Next

Retry

BCR

Feed

L.Reset

Stop

End

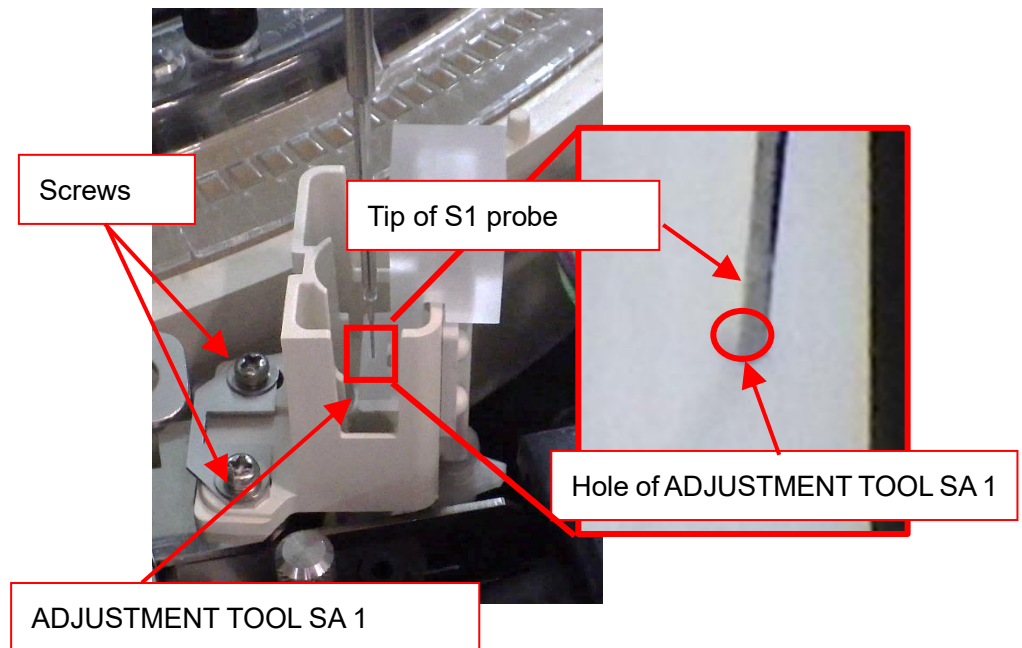
792-6427

791-1830

Procedure

(a) Adjust so that the positional relation between the probe and the rinse station is as shown in the figure.

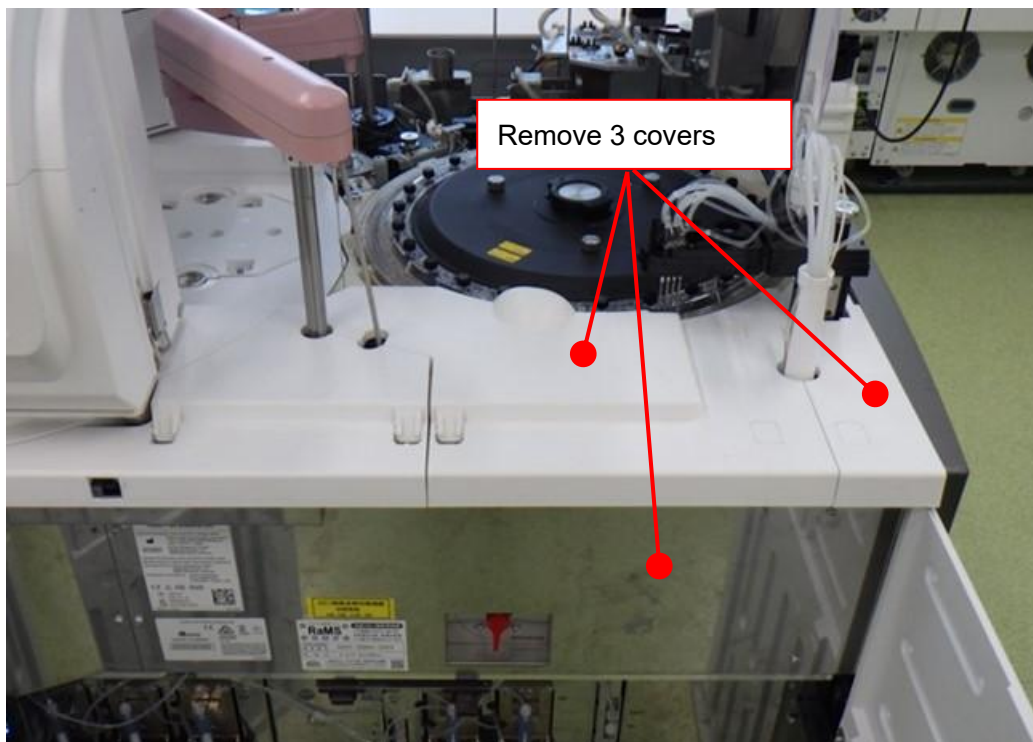
- 4.5. Shift the position from the screws (Hardware) of the CLEANING BATH S1A so that the tip of the probe enters hole of Adjustment tool. If a SPACER 10(P/N 792-1351) is placed under the S1 CLEANING BATH S1 A, tighten the screws while pressing the spacer in to prevent it from shifting.



- 4.6. Perform adjustment of the Sample Probe S1 Horizontal other than S1 Rinse station.
- 4.7. Perform adjustment of the Sample Probe S1 Vertical.

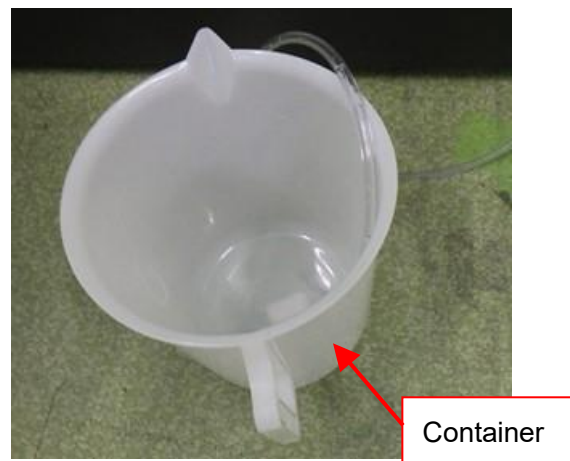
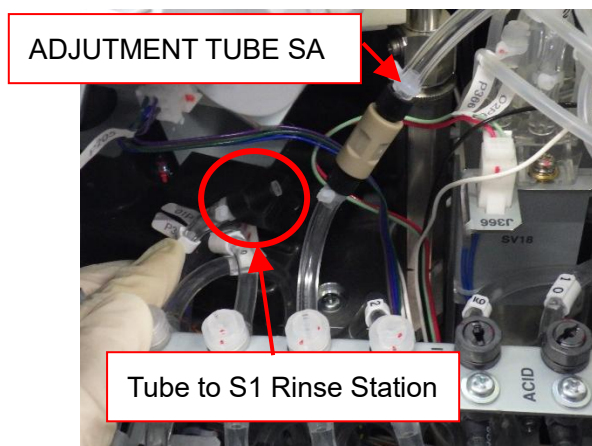
5. Adjustment of outside rinse water volume

5.1. Remove 3 covers.



5.2. Connect P/N: 792-7752 "ADJUTMENT TUBE SA".

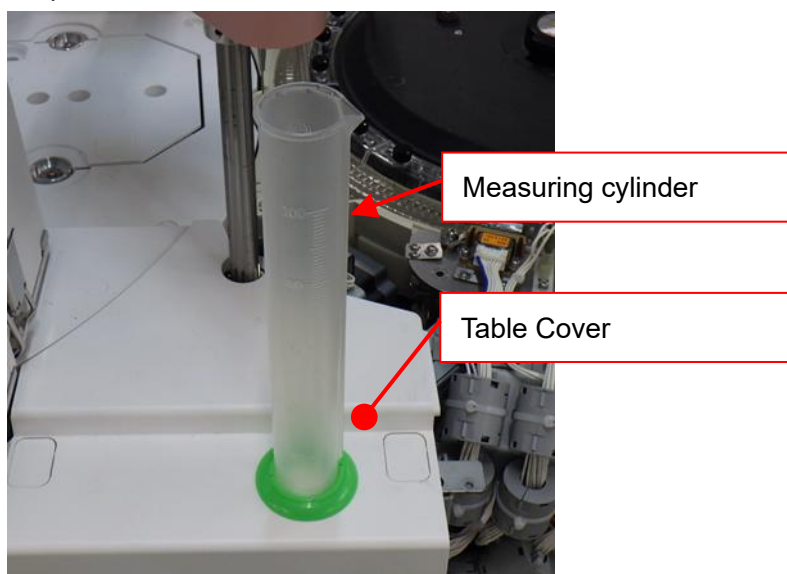
Connect one end to the needle valve and the other end to a container.



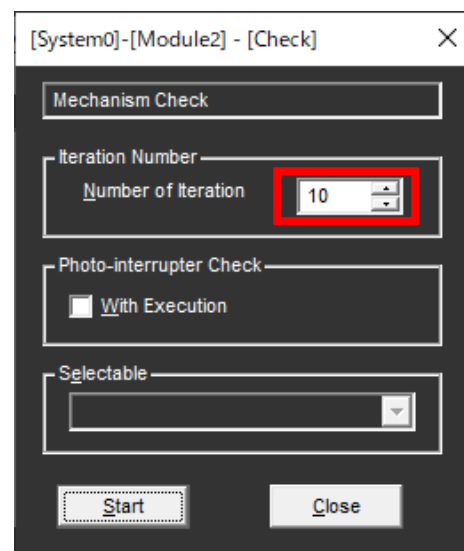
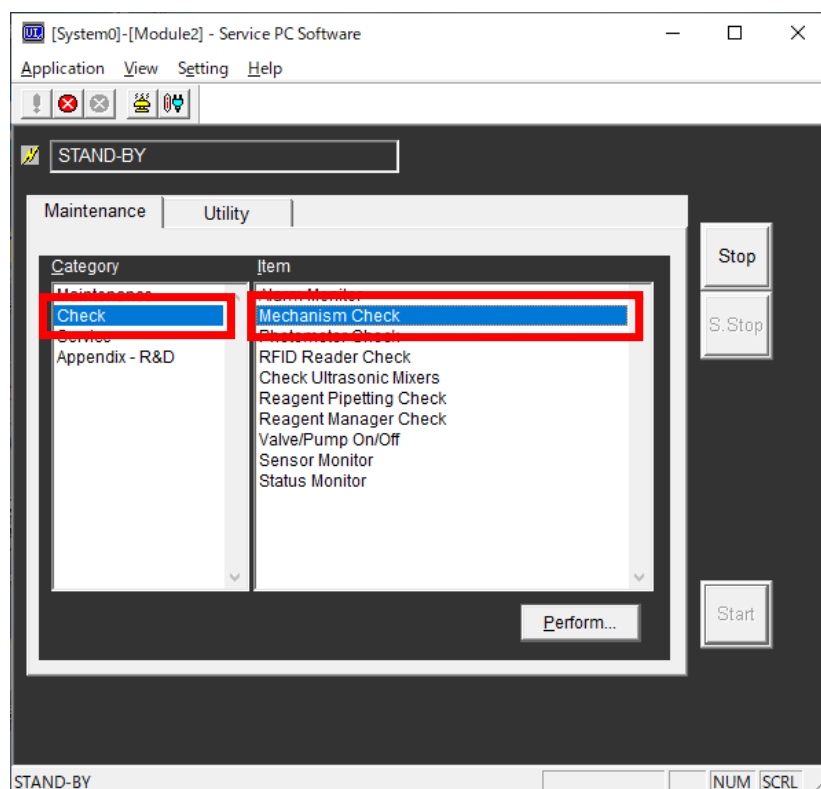
5.3. Put Measuring cylinder on Table Cover.

Be careful not to hit the R1 sampling Arm to Measuring cylinder on mechanism check.

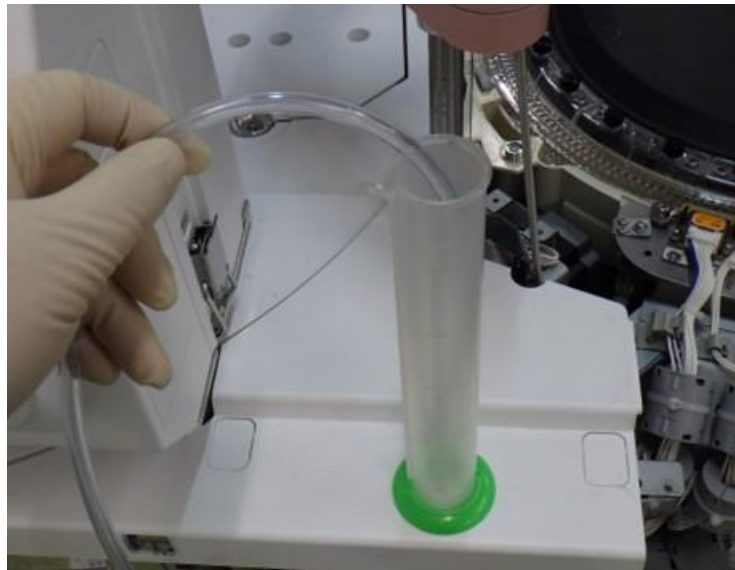
Do not put Measuring cylinder on another height. Because water volume is changed depending on height at which the Measuring cylinder is placed.



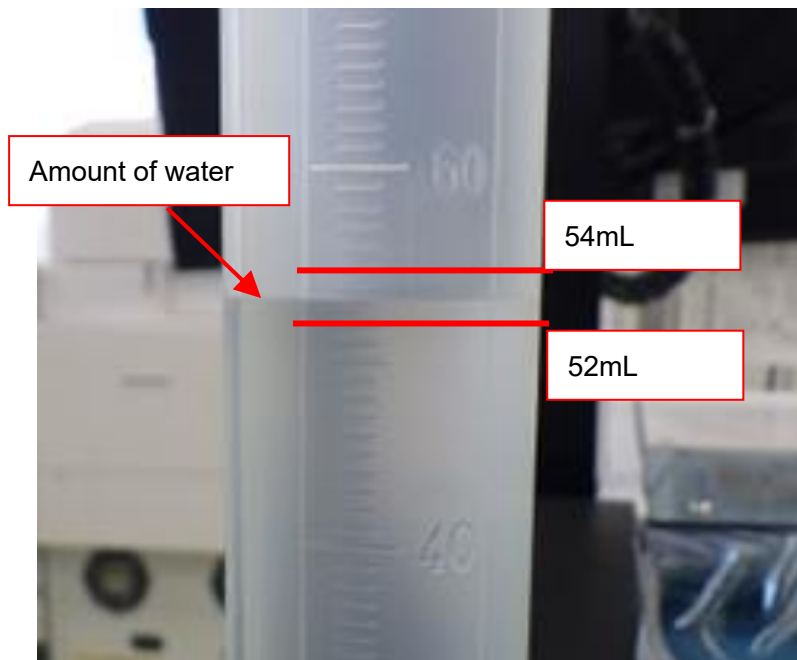
5.4. Perform mechanism check 10 times by SSW.

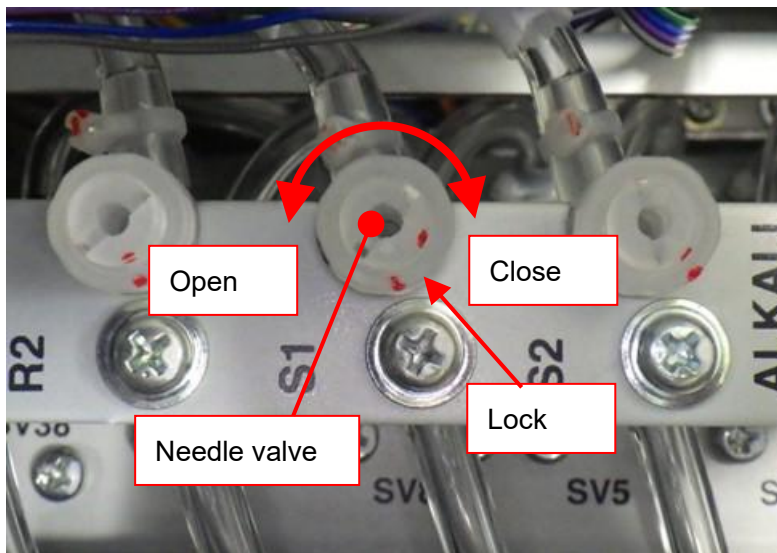


Use a Measuring cylinder to measure the amount of water after the Rinse unit starts to move with up and down.

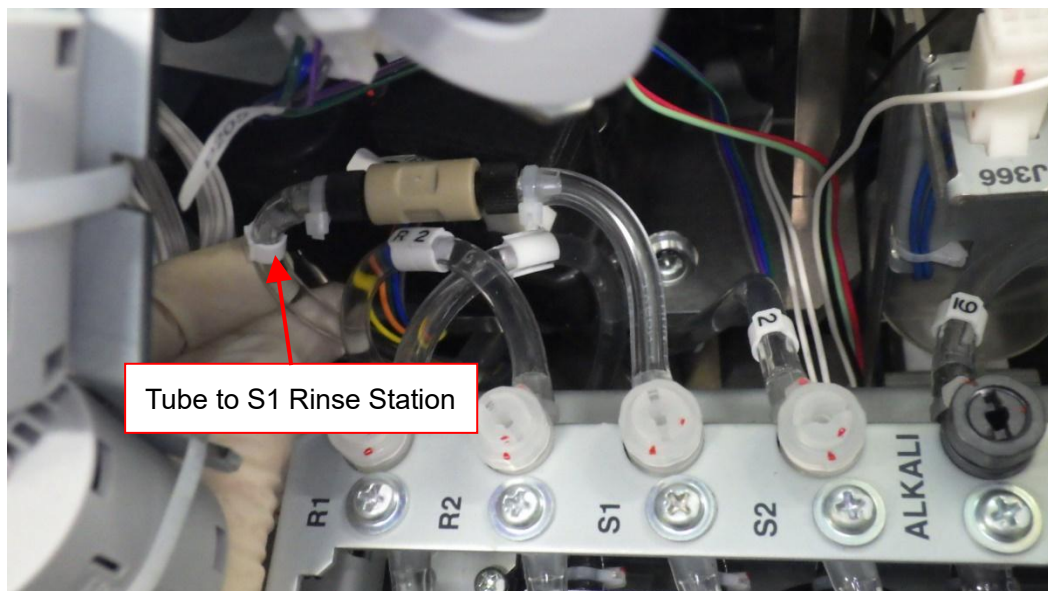


5.5. Adjust the needle valve so that the water volume is in the range of 53 ± 1 mL

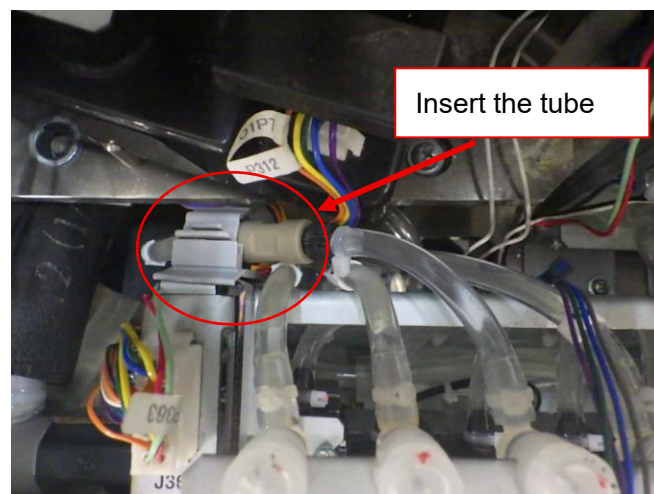




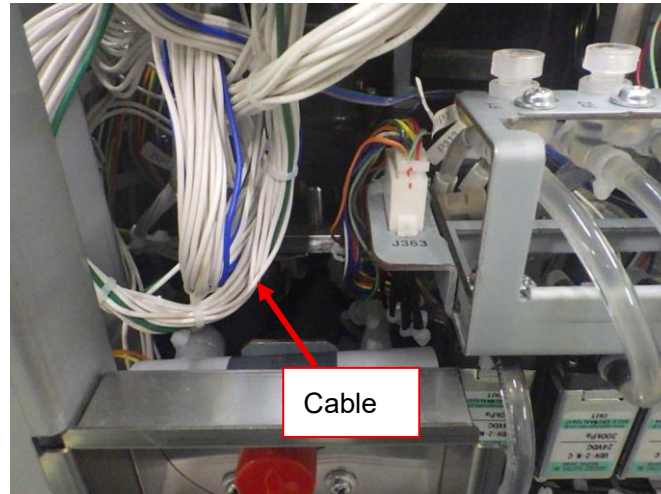
5.6. Remove the adjustment tube and reconnect the S1 Rinse Station tube.



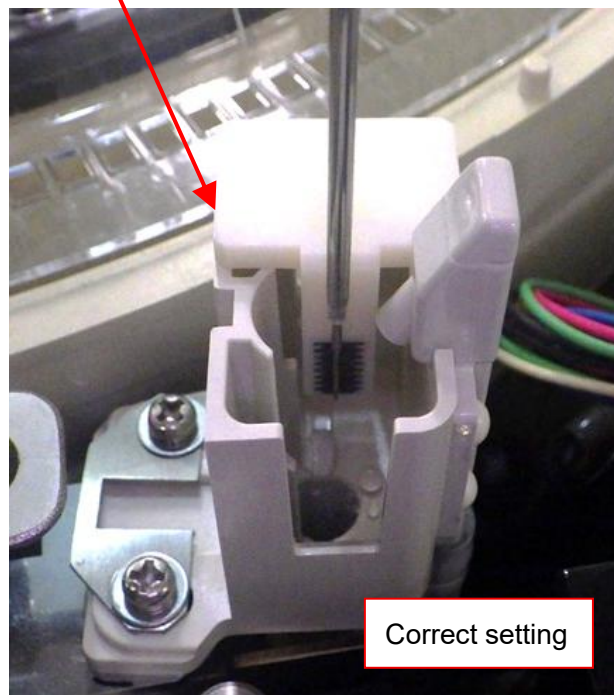
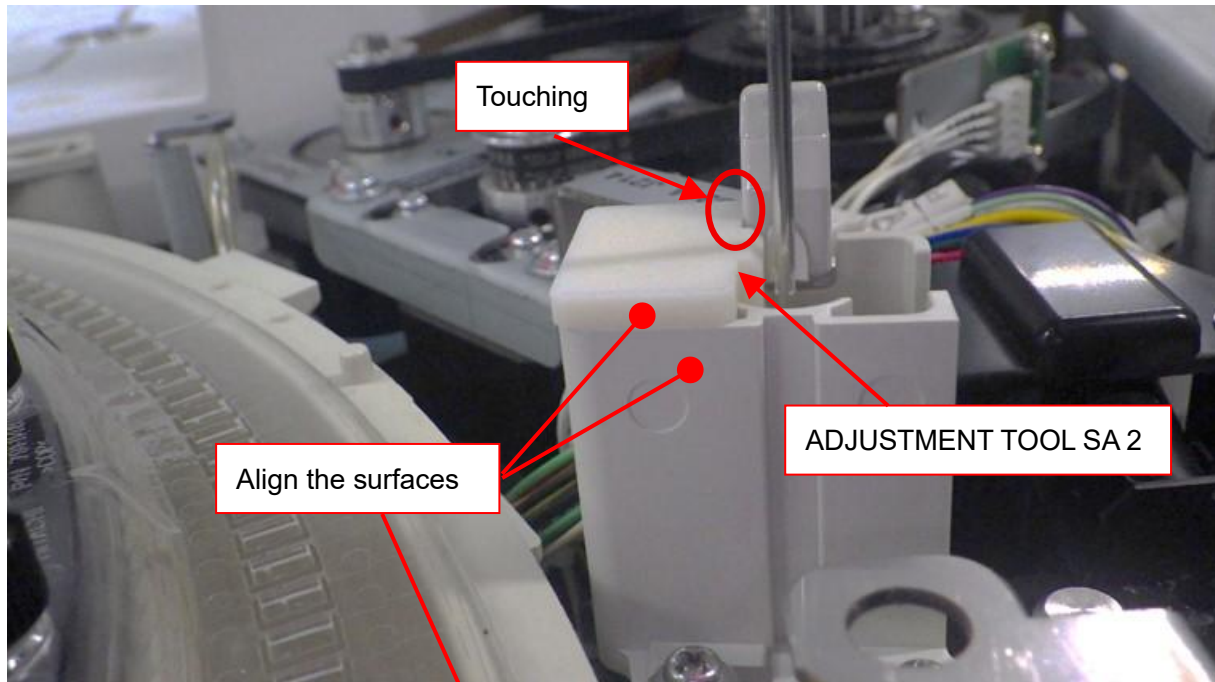
Insert the tube to the position shown in the diagram below.

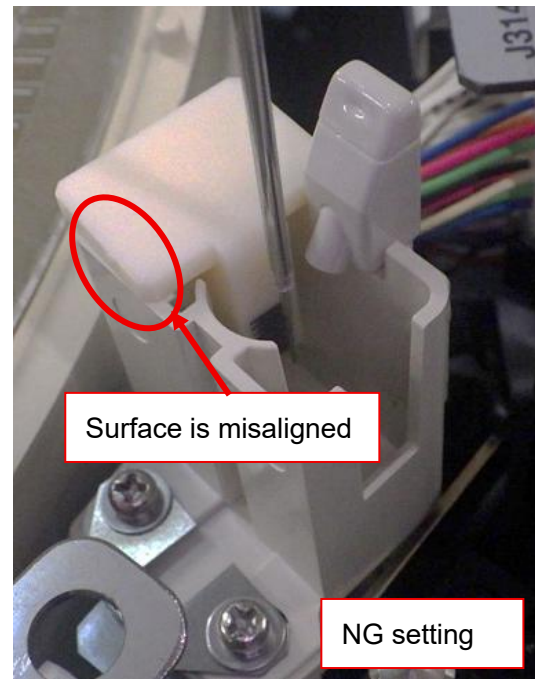
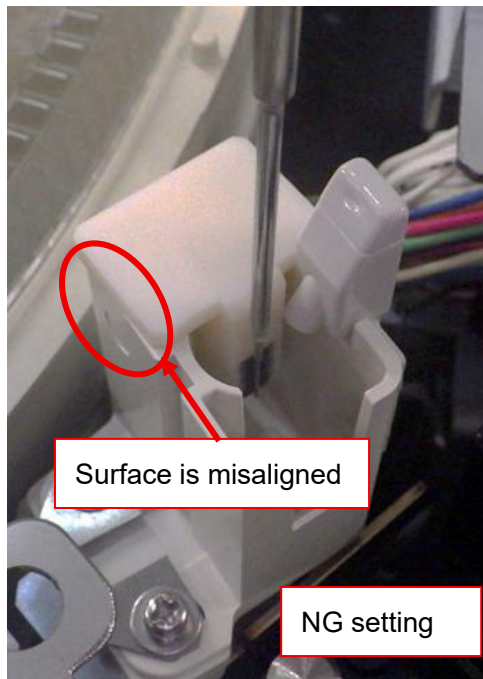


Make sure the cable is not pushing against the tube.

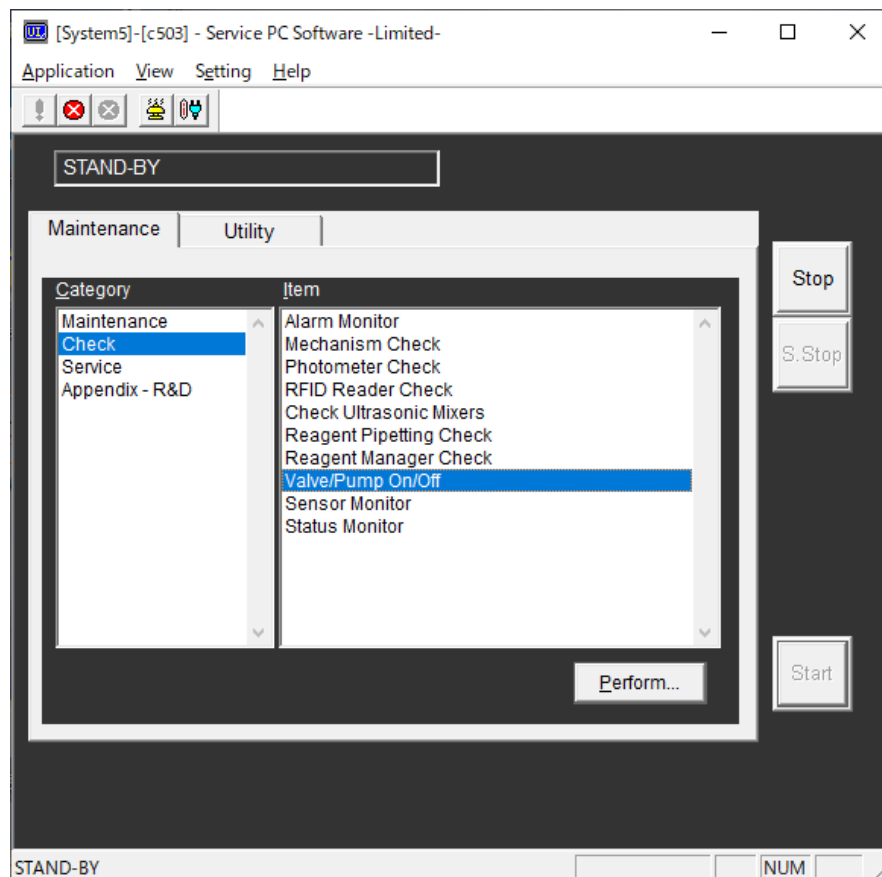


6. Make sure that the cleaning range of S1 probe.
- 6.1. Put ADJUSTMENT TOOL SA 2 on S1 Rinse station.

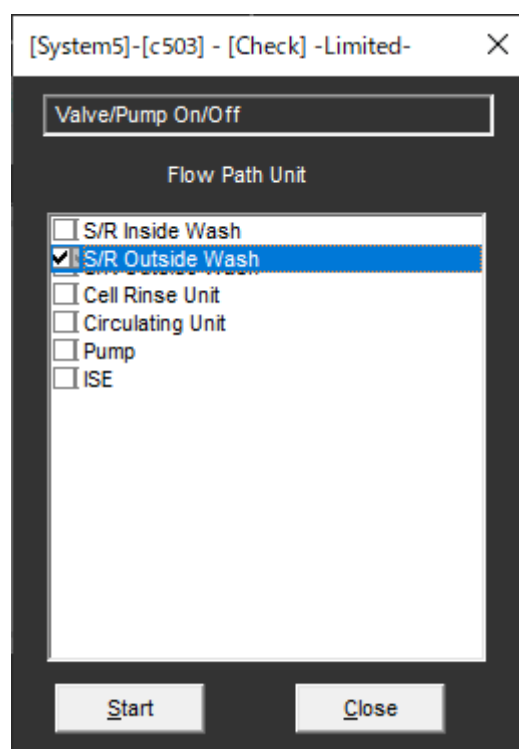




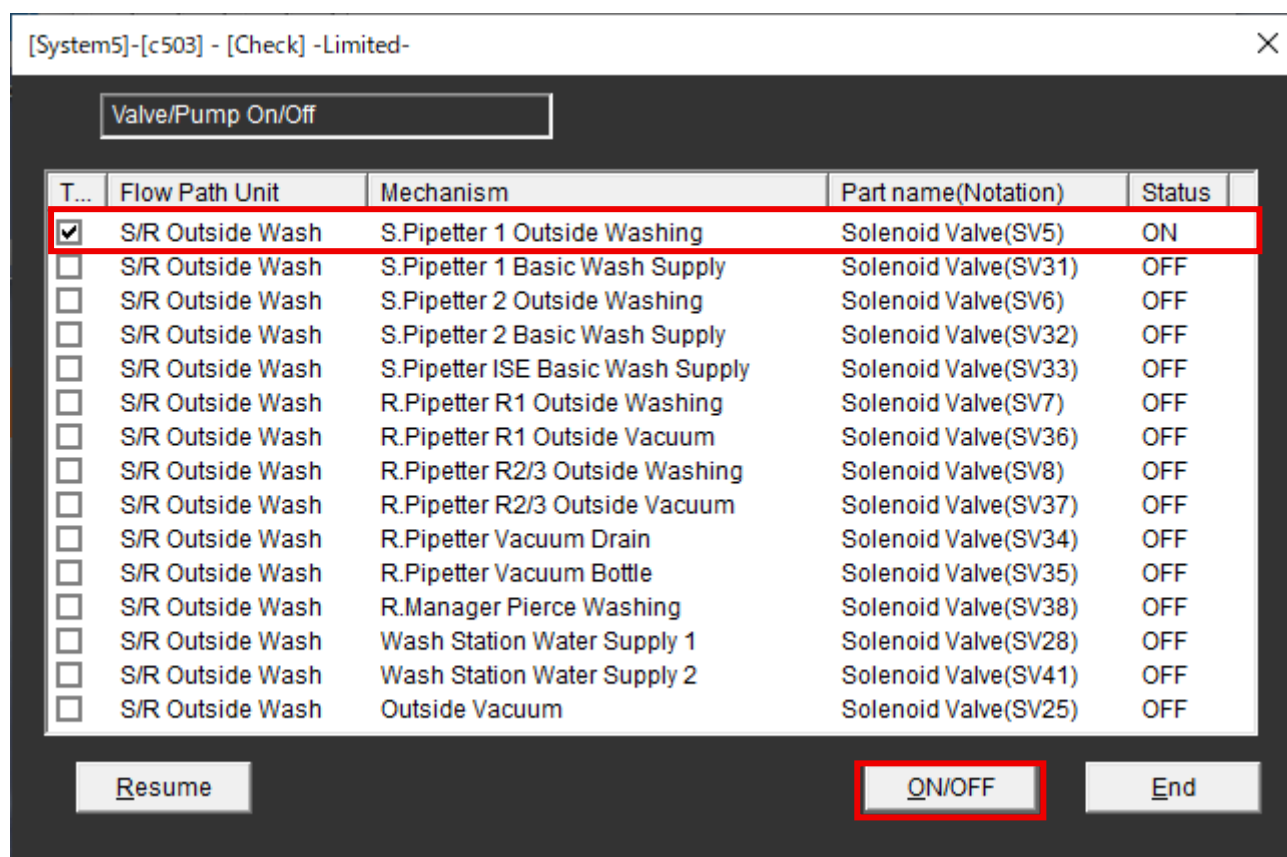
6.2. Perform "Valve/pump on/off" from SSW



6.3. Select "S/R Outside Wash"



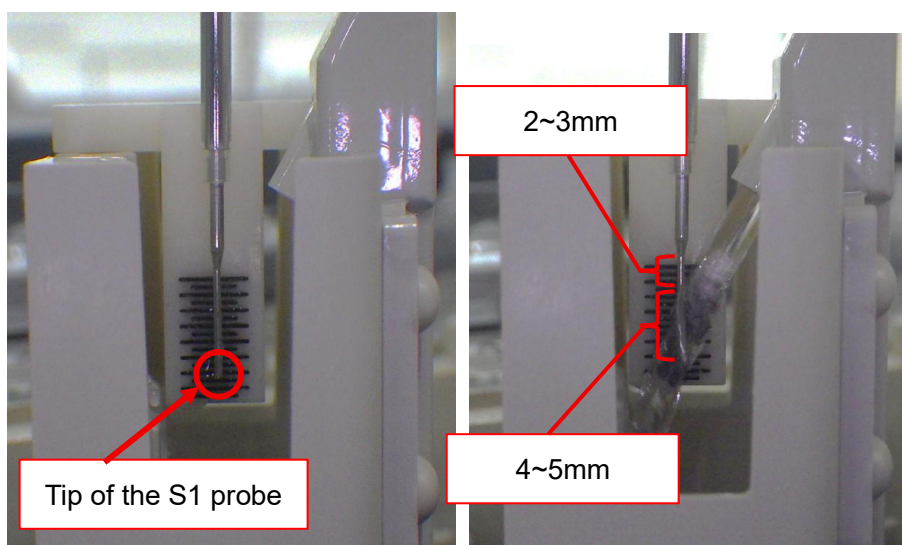
6.4. Check "S. Pipetter 1 Outside Washing" and push ON/OFF button to the status on.



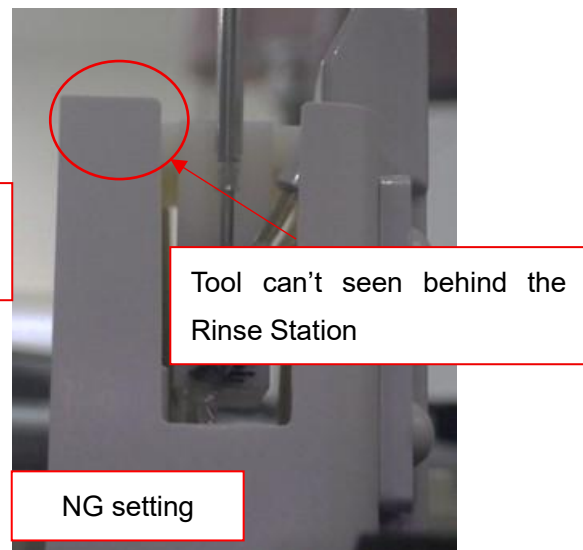
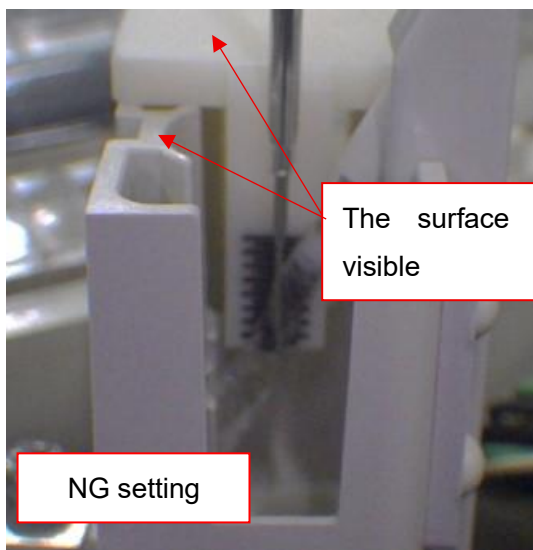
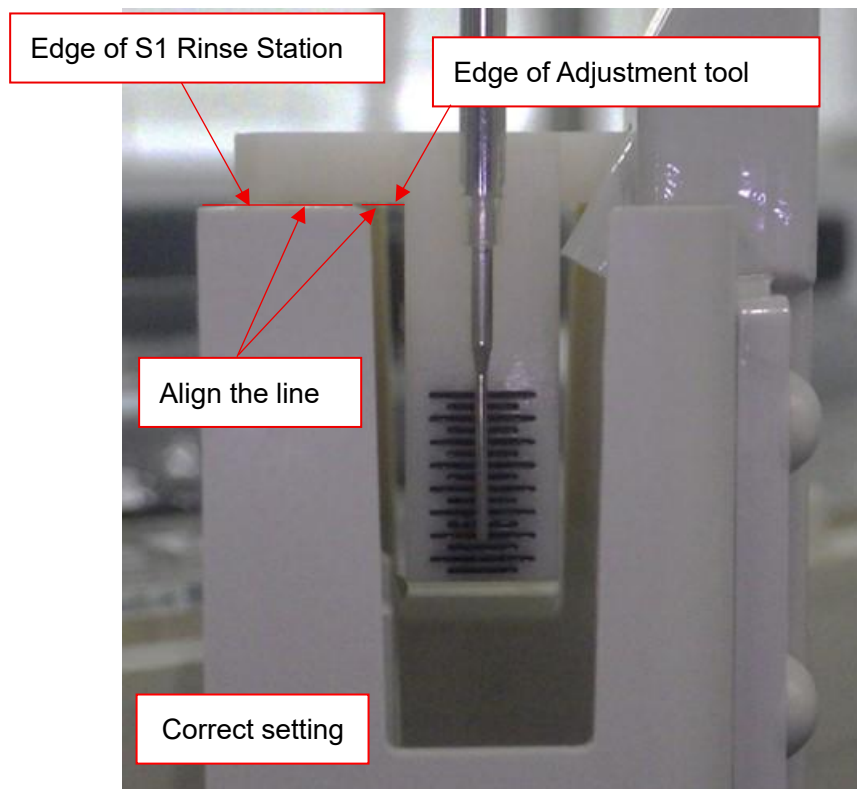
- 6.5. Make sure that the cleaning range on the left side of the probe is from 4 to 5 mm. Or the range where the outside washing water does not hit from the squeeze area is 2-3 mm. Please make sure whichever is easiest to see range.

If it is difficult to see the cleaning range, make sure the cleaning range by taking a picture with a camera.

When taking pictures with a camera, take 3 shots and make sure they are within cleaning range all 3 times. If it is difficult to see the tip of the S1 probe, take pictures with and without outside rinse water is coming out and compare them.



When taking the picture, make sure the camera and S1 probe are parallel position.



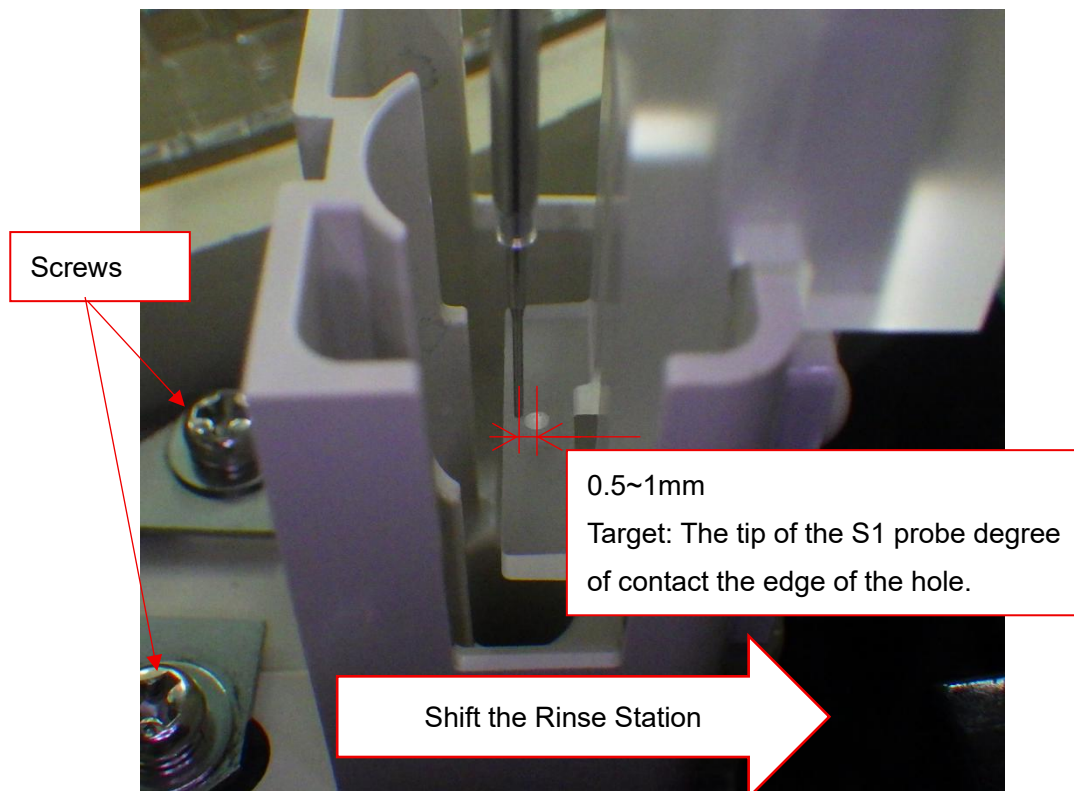
6.6. If it is out of range, readjust it using the following method.

After readjustment, check the cleaning range again.

a) If the cleaning range is less than 4mm, or the range not hit with outside washing water is exceed 3mm return to step 4.

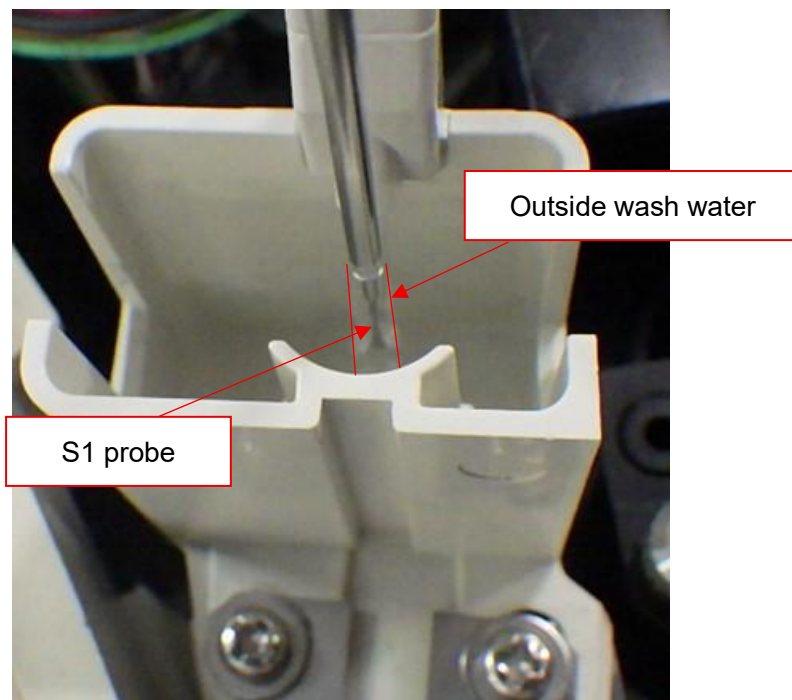
b) If the cleaning range exceeds 5mm, or the range not hit with outside washing water is less than 2mm.

b-1) Loosen the screws and shift the Rinse Station that the tip of the S1 probe degree of contact the edge of the hole.



b-2) Perform steps 6.2 to 6.4 to discharge the Outside wash water by "valve pump on/off".

b-3) Make sure the S1 probe is in the center of the Outside wash water.



b-4) Return to step 6.1.

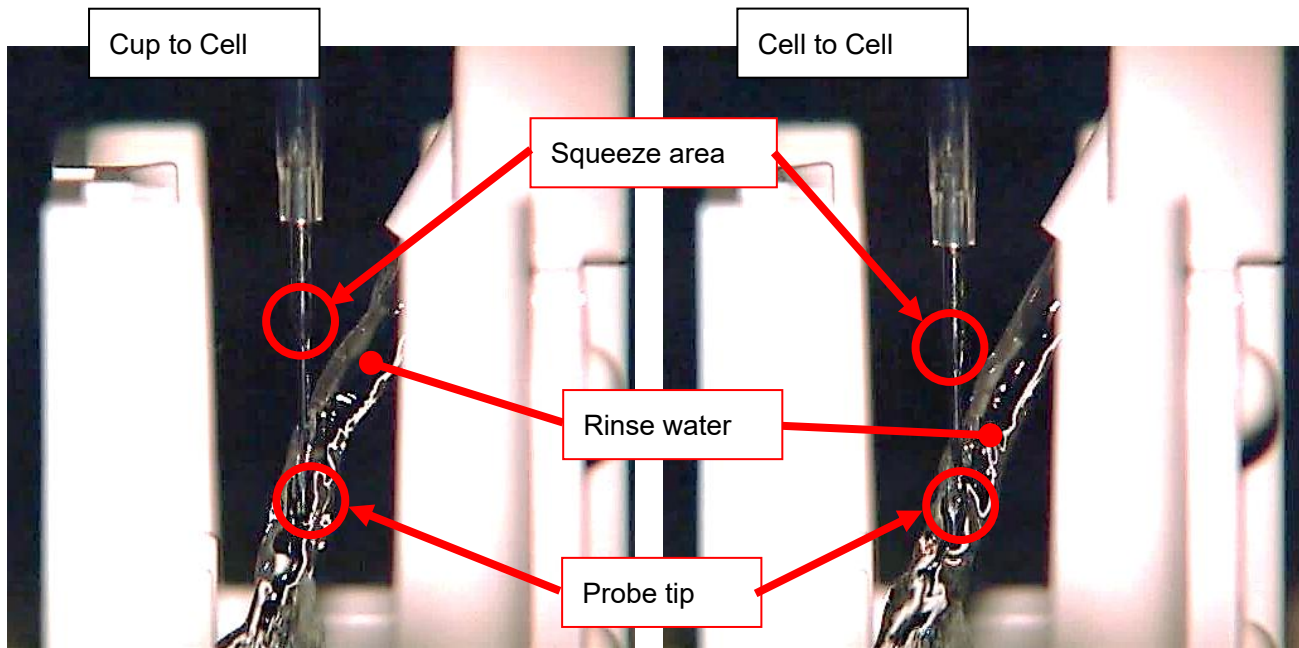
6.7. Make sure the cleaning range by the Mechanism check.

The probe descends 1 mm more at "Cell to Cell" than at "Cup to Cell" movement.

Make sure that rinse water doesn't splash squeeze area.

Make sure that tip of probe doesn't protrude from the rinse water.

If the rinse water splash to squeeze area, or tip of probe protrude from the rinse water recheck cleaning range.



7. Check and adjust water volume of Rinse unit. Refer to the following section of the service manual for adjustment procedures.

Instrument components / Fluidics / Water volume / Check / adjust / Adjusting the cell rinse water volume

8. Put back the removed cover

End